

Parameters for LED characterisation

- Total light output
- Spectral radiant flux
- Spectral luminous flux
- Colour (chromaticity)
- CCT (correlated colour temperature)
- Spectrum
- Angular distribution
- Junction temperature
- Irradiance/illuminance
- Forward voltage
- Eye safety (ANSI RP-27, IEC 62475, CIES009)
- Lifetime

Courtesy: Orb Optronix

then on the finished product.

"Every LED chip undergoes photometry test for its optical parameters before it is used in any design. There are automated machines that capture all the features of the LEDs with computer software. When a luminaire, say, a module or a product, is made out of a series of LEDs, it is also put through the same test," shares Gaurav Arora, marketing manager, Goldwyn.

Agreeing to this, Deepak Loomba, managing director and CEO, DeCore, says, "Many people often mistake the optical output characteristics of an LED to be the final product's, which is not so. An LED when converted into a product after being enclosed in cases loses light, thereby reducing the effective lumen. So final optical testing like in case of any other lighting device should be done after LEDs are packaged as a final LED application."

LED driver is an important electronic component as it interfaces with the actual power supply. "All the electrical tests like HV test, electromagnetic interference test, electromagnetic compatibility test and surge test related to electrical functioning of the product are performed on the LED drivers." After the LED driver, it's the filtered and clean power which is fed to the LED chip. Hence quality and selection of LED drivers matter a lot.

"In electrical tests, luminous-inten-

T&M Equipment for LEDs and LED products

- Radiometer
- Photometer
- Goniometer
- Integrating sphere
- LED profilers
- Spectrophotometer

sity-voltage (L-I-V) tests are conducted, while for optical tests the most important parameters are luminous flux (lm) for single LED or luminaire efficacy in lumen/watt, i.e., total light output divided by total electrical input power," shares Sahay.

Luminous efficacy is the most important parameter for LED manufacturers to dominate other lighting devices such as incandescent lamps and CFLs. Sahay explains, "For high-power LEDs, thermal is a very critical and most important parameter, as these are very sensitive to junction temperature. The most widely acceptable test standard IESNA LM-79 recommends LED manufacturers to report the flux of an LED at a particular temperature."

"For accurate high-power LED testing, heat-sink material properties are a must as all the light and efficiency characteristics of an LED might change

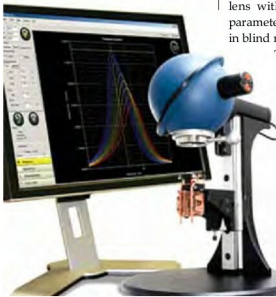
in case the heat-sink on which it is mounted changes," shares Loomba.

An ETO characterisation system measures the spectral light output of LEDs while precisely controlling electrical and thermal parameters. With today's automated test systems, you are able to select the input test parameter ranges and start the test. Advanced data acquisition and data capture capabilities provide all the necessary data to make fundamental design decisions regarding LED selection, thermal management, luminous efficacy, etc. Some of the ETO systems are capable of displaying the relative data in the form of graphs.

Goniometry for directivity parameters

LED goniometric test is used in far-field pattern and angular characterisation of an LED. Buddhadev says, "Single-chip LED goniometry performed on the chip itself in two-dimension (2-D) helps to know the angle of maximum lumens, distribution and lux level at certain distance, beam angle, etc. The goniometric test in 3-D helps in characterising the distribution of LED lumens in both X and Y axes with respect to the viewing angle. This helps to design reflector and LED lenses. Designing the reflector and lens without knowledge of angular parameters means just shoot-and-shot in blind night."

The complete product 3-D goniometry ensures that the design of luminaries is as per the requirement and it does not have any dark spot or error. There are some latest, compact, low-cost goniometer measurement assemblies optimised for fast, accurate angular measurements. These are compatible with spectro-radiometers, monochromators and photometers, making them ideally suitable for performing colour and/or intensity measurements in



ETO characterisation system